

Candidate: AXSM — AXS-05 Alzheimer Disease Agitation PDUFA (April 30, 2026)

Source: AXSM_ADA_PDUFA.md · Profile: fda_pdufa · Rendered: 2026-04-16

Candidate: AXSM — AXS-05 Alzheimer Disease Agitation PDUFA (April 30, 2026)

Score: 30.75 / 42.5 (elevated from 25.5 watchlist in Session 25, verified Session 26)

Status: ACTIVE CANDIDATE

Ticker: AXSM (Nasdaq) — Axsome Therapeutics, Inc.

Market cap: ~\$9.16B @ \$178.90 close 2026-04-09

PDUFA: 2026-04-30 (T-14 trading days from 2026-04-10)

Catalyst type: sNDA binary — first session submitted label for Alzheimer disease agitation (ADA)

Candidate file created: Session 26 (2026-04-10)

Based on: working/session25_axsm_reeval.md + Session 26 verification pull

1. Thesis Statement

What the market appears to be pricing (implicit): ~45–55% approval probability, based on modest pre-PDUFA run (+5.6% Apr 9) vs. the \$230–245 sell-side price targets.

What we estimate: 60–70% approval probability, based on:

- 3 of 4 pivotal Phase 3 trials POSITIVE on primary endpoint (ADVANCE-1, ACCORD-1, ACCORD-2)
- Priority Review granted (FDA positive directional signal)
- Backbone drug (Auvelity) already approved for MDD since 2022 — large commercial safety database exonerates major class-safety concerns
- Commercial preparation behavior (sales force doubling 300→600, acquisition of balipodect on Apr 1) inconsistent with internal CRL expectation
- No AdCom announced (verified Session 26 primary and secondary sweep)
- Unmet need is massive — only brexpiprazole (Rexulti) approved with boxed warning

Delta: ~15 percentage points of probability mispricing × ~30% upside on approval = ~4.5% expected-value mispricing above current price. Score 30.75 reflects moderate conviction tempered by genuine asymmetry risk (ADVANCE-2 miss is real, not imaginary).

2. Primary-Source Verification (Session 26)

EDGAR (CIK 0001579428)

Direct data.sec.gov/submissions pull:

| Date | Form | Item | Material? |

|-----|-----|-----|-----|

| 2026-04-01 | 8-K | 8.01 (Other Events) | **No** — balipodect acquisition announcement (Item 8.01 is informational, not regulatory) |

| 2026-02-23 | 8-K | — | Routine (pre-S25) |

| 2026-02-17 | 8-K | — | Routine (pre-S25) |

| 2026-01-12 | 8-K | — | Routine (PDUFA acceptance) |

Zero 8-Ks containing FDA action, CRL, AdCom, or label negotiation since sNDA acceptance Jan 12. Item 8.01 on the Apr 1 filing explicitly references balipodect acquisition press release — no FDA content.

openFDA Auvelity post-marketing surveillance (Session 26 live query)

Top serious reactions (commercial MDD use, 2022–2026):

- Suicidal ideation: 97
- Depression: 48
- Serotonin syndrome: 48
- Dizziness: 40
- Off-label use: 38
- Seizure: 34

Interpretation: These are known class effects for bupropion (seizure, suicidal ideation) and dextromethorphan (serotonin syndrome, dissociation). They are already in the current Auvelity label. No novel signals that would materially change the benefit-risk calculation for an ADA label. The ADA population is more frail and more elderly than MDD, which could surface different dose-response, but the aggregate AE counts are not alarming for a 3-year commercial drug.

FDA AdCom calendar (Session 26 verification)

No Psychopharmacologic Drugs Advisory Committee meeting announced for April 2026. Last PDAC meeting was July 18 2025 (Federal Register). No AXS-05 referrals in Session 26 web sweep. **Absence of AdCom is a positive marker** — brexpiprazole's 2023 ADA approval process included a fractious AdCom. FDA choosing NOT to convene AdCom for AXS-05 suggests the review division does not see the data package as requiring external expert adjudication.

Price Action

| Date | Close | Vol | Delta |

|-----|-----|-----|-----|

| 2026-03-19 | 158.88 | 455K | — |

| 2026-03-26 | 164.26 | 304K | +3.4% |

| 2026-03-31 | 169.02 | 677K | +2.9% |

| 2026-04-01 | 171.78 | 604K | +1.6% (balipodect M&A day, neutral-to-positive tape) |

| 2026-04-07 | 168.74 | 401K | -1.8% |

| 2026-04-08 | 173.19 | 525K | +2.6% |

| **2026-04-09** | **178.90** | **986K** | **+3.3% on 2× volume** |

| 2026-04-10 | 178.11 | 560K | -0.4% (held gains, normal vol) |

20-day drift +12.3%. The Apr 9 session is the standout — 2× average volume without a known catalyst. Most likely an institutional accumulation before PDUFA window closes. Apr 10 held the gains on normal volume — no distribution. **Enters T-14 intensive window Apr 15 (S44 confirmed).**

S44 Analyst Update (Apr 14, 2026)

- **UBS** (Ashwani Verma): Maintained Buy, raised PT to **\$259** (Apr 10) — above prior consensus
- **Consensus:** 16 analysts, Buy, avg PT \$197.95
- **H.C. Wainwright:** Previously raised PT on Priority Review
- **No downgrades detected** — kill condition #6 remains CLEAR

S44 Insider Activity Update

- CEO Herriot Tabuteau: \$6M options exercise (Feb, 10b5-1 planned) — neutral
- Director Mark Coleman: 6,250 option exercise at \$4.95, HELD shares (Feb 23) — mildly bullish
- **No insider selling detected** — positive for thesis

3. Clinical Evidence Base

AXS-05 is a fixed-dose combination of dextromethorphan (an uncompetitive NMDA receptor antagonist and sigma-1 agonist) and bupropion (a CYP2D6 inhibitor that boosts dextromethorphan plasma concentrations and a weak NDRI itself). Already FDA-approved as Auvelity for major depressive disorder (2022).

ADA Phase 3 Program — 4 Trials

| Trial | Design | Result on CMAI | Status |

|-----|-----|-----|-----|

| **ADVANCE-1** | RCT, AXS-05 vs placebo, ~400 pts, 5 weeks | **POSITIVE** — stat-sig reduction vs placebo | Topline reported 2022 |

| **ADVANCE-2** | RCT, 408 pts | **MISSED primary** — 13.8 vs 12.6 placebo (small placebo delta) | Topline reported 2023 |

| **ACCORD-1** | Relapse prevention, responder randomization | **POSITIVE** — stat-sig longer time to relapse | Published Neurology 2024 |

| **ACCORD-2** | Second relapse prevention trial | **POSITIVE** — presented AAN 2025 | Apr 2025 |

Hit rate: 3 of 4 on primary endpoints. The FDA has approved drugs on less — lecanemab was approved on 1 positive Phase 3 (CLARITY-AD). What matters for the review division is:

1. Do the positive trials triangulate a consistent effect size? (Yes — ADVANCE-1, ACCORD-1, ACCORD-2 all significant)
2. Is the miss explainable? (ADVANCE-2 had an unusually large placebo response — 12.6 point reduction vs. placebo norms of ~8–10; this is a known issue in agitation trials)
3. Is the effect clinically meaningful? (Cohen's d effect sizes in positive trials were moderate, clinically meaningful per FDA agitation guidance)

The "mixed program" framing (Session 24 inherited) was incorrect framing. A 3-of-4 hit rate with 2 relapse-prevention trials both positive is a stronger package than typical.

Safety — The Real Review Question

The FDA's primary concern in an ADA label is whether a new drug will increase:

- Mortality (brexpiprazole had to address this)
- Sedation/falls (fragile elderly population)
- Cognitive decline (confound the underlying disease)
- Cerebrovascular events (antipsychotics in dementia have boxed warnings)

AXS-05's advantage: It is NOT an antipsychotic. The mechanism (dextromethorphan + bupropion) has been in humans for decades with well-characterized safety. The MDD commercial dataset is large (>100K patient-years). FDA has already negotiated the MDD label.

Risks to thesis: The elderly AD population has different pharmacokinetics, and bupropion seizure risk may be elevated in frail elderly. But this is a dose issue, not a mechanism issue — addressable with labeling.

4. Commercial Setup

Auvelity MDD baseline: ~\$600M annualized run-rate (from 2025 Q4 guidance). Growing franchise with 300-rep commercial team. Already profitable at gross margin level; company guiding toward corporate profitability as Auvelity scales.

ADA opportunity:

- ~6.5M people with Alzheimer disease in US
- ~50% experience agitation at some point → ~3.25M addressable
- Current SoC: off-label antipsychotics (with boxed warnings), benzos (contraindicated in elderly), brexpiprazole (approved, ~\$180/month, ~\$1.2B peak est.)
- AXS-05 peak sales estimates: \$1.5–3.0B per analyst models

Pre-PDUFA commercial prep:

- Doubling sales force from 300 to 600 reps (stated publicly Feb Oppenheimer Conference)
- April 1, 2026: Acquired balipodect (PDE10A inhibitor for schizophrenia) — pipeline expansion
- Q1 2026 earnings scheduled May 4 (post-PDUFA) — guidance will bake in ADA assumption
- Zero layoff rumors, zero CFO departures, zero R&D cuts — behavior consistent with management expecting approval

Commercial prep is not a guarantee of approval but is strongly incentive-aligned with insider knowledge. Doubling a specialty sales force has ~\$60–100M annualized cost commitment that is very hard to unwind if CRL arrives.

5. 7-Dimension Score (Session 25 → Session 26 verified)

Dimension	Weight	Score	Rationale
Signal Strength	2.0	4.0	3-of-4 P3 hits + Priority Review + commercial prep + Apr 9 volume tape
Catalyst Clarity	1.0	4.5	Hard PDUFA April 30, binary
Information Asymmetry	1.5	2.5	Market implied ~50% vs our ~65% — real but modest edge
Risk/Reward	1.0	3.5	Upside +20–30% (Jefferies PT \$245), downside –25–35% (MDD franchise floors)
Edge Decay	1.0	3.0	14 trading days — will compress
Liquidity	1.0	4.0	\$9.16B cap, 500K–1M ADV
Catalyst Timeline	1.0	4.0	Known hard date

Weighted total: $8.0 + 4.5 + 3.75 + 3.5 + 3.0 + 4.0 + 4.0 = 30.75 / 42.5$

6. Kill Conditions (explicit, monitor daily)

| # | Condition | Source | Action |

|---|-----|-----|-----|

| 1 | FDA AdCom announced for AXS-05 | Federal Register + AXSM 8-K | Immediate archive + lessons memo |

| 2 | Price break <\$165 | yfinance | Score -3 (de-rating signal); reassess |

| 3 | Price break <\$155 | yfinance | Instant archive; thesis broken |

| 4 | New safety signal in openFDA Auvelity commercial data | api.fda.gov | Score -2; reassess |

| 5 | CRL or delay announcement before April 30 | SEC 8-K | Instant archive; lessons memo |

| 6 | Sell-side downgrade to <\$200 PT from Jefferies, Mizuho, Morgan Stanley, Goldman | News search | Score -2; reassess |

| 7 | Short-interest spike >15% float or unusual put buildup | FINRA/OCC data | Score -1; reassess |

Session 26 status: All kill conditions CLEAR.

7. PDUFA Day Protocol (April 30, 2026)

T-14 to T-1: Daily kill-condition sweep. EDGAR 8-K monitoring. FDA press release monitoring. Option chain IV check for implied move.

T-0 (April 30):

- 09:00 UTC: Final 8-K pull
- 13:30 UTC (9:30 ET): Market open — observe gap direction
- Throughout session: FDA CDER press release monitoring at <https://www.fda.gov/news-events/press-announcements>
- 16:00–22:00 UTC: Peak decision announcement window (FDA typically announces 4–6pm ET)

Outcome handling:

- **Approval:** Update thesis, monitor label specifics (dose, warnings, REMS), reassess score. Expected price move +15–30%.
- **CRL:** Archive immediately. Write lessons-learned memo. Expected price move -25–40%.
- **Delay/PDUFA extension:** Rare for Priority Review; would itself be material. Instant deep investigation.

8. Position Policy

This is an **ACTIVE CANDIDATE for awareness**, not a recommendation to enter a new position at \$178.90. Risk management:

- If flat going in: **HOLD**, no new entry at current level. Entry edge has compressed.
- If long from <\$170: **HOLD** through decision. Kill conditions clear.
- If long from <\$160: Consider trimming 25% to protect accumulated P&L. Kill conditions clear, but decision-day vol is real.
- If long from >\$180: Caution. Upside compressed.

This system does not execute trades. Outputs are analytical only.

9. What Would Move the Score

+2–3 points:

- Additional sell-side upgrade with PT >\$260
- Option chain implied move narrowing significantly (market getting more confident)
- Insider buying disclosed (not in S25, not in S26)
- New primary-source confirmation that AdCom waived formally

–3 to –5 points:

- Any 8-K mentioning FDA communication
- Any AdCom announcement
- Price break below \$165
- Sell-side downgrade

10. Cross-References

- `working/session25_axsm_reeval.md` — original re-score documentation
- `working/session26_axsm_verification.md` — TBW if deeper primary pulls needed next session
- `candidates/TVTX_FSGS_PDUFA.md` — comparable active PDUFA candidate
- `candidates/VRDN_veligrotug_PDUFA.md` — comparable active PDUFA candidate (June)
- `framework/scoring_system.md` — 7-dimension rubric definition
- `DECISIONS.md` D-033 — elevation decision rationale

11. Sources

- [FDA Grants Priority Review AXS-05 for ADA (PDUFA 2026-04-30)](<https://www.hcplive.com/view/fda-accepts-grants-priority-review-to-axs-05-snda-for-alzheimer-s-disease-agitation>)
- [Axsome P3 Program Successful Completion (Dec 2024)](<https://www.globenewswire.com/news-release/2024/12/30/3002588/33090/en/axsome-therapeutics-announces-successful-completion-and-results-of-phase-3-clinical-program-of-axs-05-in-alzheimer-s-disease-agitation.html>)
- [Axsome ACCORD-2 Results AAN 2025](<https://www.globenewswire.com/news-release/2025/04/04/3055865/0/en/Axsome-Therapeutics-Presents-Positive-Pivotal-Phase-3-Results-of-the-ACCORD-2-Trial-of-AXS-05-in-Alzheimer-s-Disease-Agitation-and-the-SYMPHONY-Trial-of-AXS-12-in-Narcolepsy-at-the.html>)
- [Axsome Balipodect Acquisition Apr 1 2026 8-K](<https://www.sec.gov/Archives/edgar/data/1579428/000119312526135669/axsm-20260401.htm>)
- [Axsome Q1 2026 Earnings Date (May 4)](<https://www.manilatimes.net/2026/04/08/tmt-newswire/globenewswire/axsome-therapeutics-to-report-first-quarter-2026-financial-results-on-may-4/2316234>)
- [Neurology ACCORD Relapse Prevention](<https://www.neurology.org/doi/10.1212/WNL.0000000000205634>)

Created Session 26 (2026-04-10). Updated Session 44 (2026-04-14): Price action, analyst, insider updates. Kill sweep #19 CLEAR. T-14 intensive begins S45.

Session 48 T-17 Intensive Deep-Check (2026-04-13 ~23:00 UTC)

Price action (5-day): \$167.90 → \$168.74 → \$173.19 → \$178.90 → \$178.11. Strong upward momentum (+6% week). Volume spike Apr 9 (986K vs avg ~500K).

EDGAR filings since last check: No new 8-Ks since Apr 1 (balipodect acquisition). Most recent: Form 4s from Feb 27, Schedule 13G/A Mar 26.

openFDA Auvelity AE: Normal cadence through end-2025 data (latest bucket: Dec 31, 2025 — 6 reports). No spike or novel signal.

Options IV: May 15 expiry (brackets PDUFA): ATM straddle ~\$33.5 on \$178 = ~18.8% implied move. Put OI concentrated at \$180-185 strikes. April 17 expiry: minimal, pre-PDUFA.

Analyst sentiment: Unchanged — Strong Buy consensus, 16B/1H/OS, avg PT \$226.29 (27% upside). UBS PT \$259 (S45 note).

Insider activity: CEO \$6M option sale in March via pre-planned 10b5-1. No new Form 4s since Feb 27.

News: Axsome highlighted at Oppenheimer conference (Feb 28) — Auvelity momentum, 600-rep salesforce deployment, April 30 ADA PDUFA. Q1 2026 earnings scheduled May 4 (4 days post-PDUFA). Company acquired balipodect (PDE10A inhibitor) from Takeda Apr 1 for schizophrenia pipeline — signals management confidence and capacity for deal-making ahead of binary event.

Kill condition sweep #21: ALL CLEAR.

- #1 AdCom: No PDAC meeting announced. CLEAR.
- #2 Price <\$165: \$178.11. CLEAR.
- #3 Price <\$155: CLEAR.
- #4 Safety signal: No novel AEs. CLEAR.
- #5 CRL/delay: No 8-K. CLEAR.
- #6 Downgrade: No sell-side downgrades. CLEAR.
- #7 SI spike: 6.4% (stable). CLEAR.

Session 49 T-16 Intensive Deep-Check (2026-04-14 ~00:15 UTC)

Price action (5-day): \$168.74 → \$173.19 → \$178.90 → \$178.11 → \$175.72. Slight pullback from \$178.90 peak (-1.8% from Apr 9 high). Volume on Apr 13: 174,652 (below avg ~607K — light trading day, likely post-tax-day effect). Weekly range: \$167.14–\$182.50.

EDGAR filings since last check: No new filings since Apr 7 (Form 4). Most recent 8-K still Apr 1 (balipodect acquisition). No CRL-related communications.

openFDA Auvelity AE: 2,727 total reports through latest data. Most recent bucket: Dec 31, 2025. One Stevens-Johnson syndrome case (serious=1) — single isolated case in 2,727 total reports is not a safety signal. No cluster, no novel pattern.

News: Search confirms positive narrative — \$155M Auvelity Q1, salesforce doubled to 600 reps, preparing for ADA launch. No negative analyst coverage, no unexpected regulatory communication.

SI: 6.4% (unchanged from S48).

T-minus: T-16 trading days (Apr 13 → Apr 30). Entering second week of intensive monitoring window.

Kill condition sweep #22: ALL CLEAR.

- #1 AdCom: No PDAC meeting announced. CLEAR.
- #2 Price <\$165: \$175.72. CLEAR.
- #3 Price <\$155: CLEAR.
- #4 Safety signal: No novel AEs. Single SJS case isolated. CLEAR.
- #5 CRL/delay: No 8-K. CLEAR.
- #6 Downgrade: No sell-side downgrades. CLEAR.
- #7 SI spike: 6.4% (stable). CLEAR.

Session 61 T-15 Intensive Deep-Check (2026-04-15 10:30 UTC)

Price: \$184.18 (unchanged vs Apr 14 close). Mcap \$9.42B. 14-day trajectory very constructive: \$161.78 on Mar 25 → \$184.18 today (+13.9% in 20 trading days). Higher highs established Apr 8-14. Volume 820K on Apr 14 vs ~500K 14d avg (suggests institutional accumulation).

Filings since last check: NONE. Last 8-K remains Apr 1. Last 13G/A Mar 26 (institutional). No Form 4 insider selling. No material corporate event.

T-minus: T-15 calendar days (Apr 15 → Apr 30).

Kill condition sweep #23: ALL CLEAR.

- #1 AdCom: No PDAC meeting announced. CLEAR.
- #2 Price <\$165: \$184.18. CLEAR.
- #3 Price <\$155: CLEAR.
- #4 Safety signal: No novel AEs in body of any recent filing. CLEAR.
- #5 CRL/delay: No 8-K. CLEAR.
- #6 Downgrade: No sell-side downgrades flagged. CLEAR.
- #7 SI spike: stable. CLEAR.

Interpretation: Price action + filing quiet + Priority Review active + no AdCom = base-case approval probability rising. Market appears to be pricing in a positive outcome. No new risks identified.

Next session: T-14 deep check (Apr 16).

Scoring Rubric — fda_pdufa

Rubric not available for this profile.